COVID-19 DATA VISUALIZATION

(Data Analytics & Business Intelligence)

Domain: Data & Business Intelligence / Data Analytics **Department:** Computer Science and Engineering (Artificial Intelligence and Machine Learning)

ALGORITHM AND DATASET

Algorithm includes:

- **Step-by-step procedure**
- **Logic or method used**
- **Model or technique applied**
- **Process flow of the system**

Dataset includes:

- **Source of data**
- **Number of records/samples**
- **Features or attributes (columns)**
- **Training and testing data split**
- **Data format or type**

1. Algorithm

The COVID-19 Data Visualization project follows a structured data analytics and Business Intelligence workflow. Raw COVID-19 records are collected, checked, cleaned and transformed into an analysis-ready dataset. Descriptive analysis and KPI calculations are then performed, and the results are presented through charts, summary indicators and an interactive dashboard. The process is designed to make large COVID-19 datasets easier to understand and compare.

1.1 Step-by-Step Procedure

Step	Stage	Procedure
1	**Data Collection**	Obtain the selected COVID-19 dataset in CSV or Excel format.
2	**Data Loading**	Load the dataset using Python/Pandas or the selected BI platform.
3	**Data Understanding**	Inspect columns, data types, dates, locations and COVID-19 indicators.
4	**Data Validation**	Check missing values, duplicates, invalid values and inconsistent formats.
5	**Data Cleaning**	Remove duplicates, handle missing values and standardize location names.
6	**Data Transformation**	Create date fields such as day, month, quarter and year where required.
7	**Data Analysis**	Calculate totals, trends, comparisons and summary statistics.
8	**KPI Calculation**	Calculate confirmed cases, deaths, recoveries, active cases and other supported indicators.
9	**Visualization**	Create line charts, bar charts, KPI cards and maps where applicable.
10	**Dashboard Development**	Combine visualizations into an interactive dashboard with filters.
11	**Validation**	Cross-check dashboard calculations against the cleaned dataset.
12	**Insight Generation**	Interpret trends, geographical differences and important patterns.

1.2 Logic or Method Used

The project uses a descriptive analytics methodology: **Data Collection** → **Data Understanding** → **Data Cleaning** → **Data Transformation** → **Data Analysis** → **KPI Calculation** → **Visualization** → **Interactive Dashboard** → **Insights.** Users can filter the dashboard by date and available geographical or categorical fields.

COVID-19 Data Visualization

1.3 Model or Technique Applied

- **Data Cleaning:** Missing-value handling, duplicate removal and consistency checks.
- **Data Transformation:** Date extraction, standardization and derived analytical fields.
- **Descriptive Analytics:** Totals, trends, comparisons and summary statistics.
- **Business Intelligence:** KPI cards, filters, slicers and dashboard-based reporting.
- **Data Visualization:** Line charts for trends, bar charts for comparisons and maps where geographical data is available.
- **AIML Extension:** The cleaned dataset can later support forecasting or anomaly detection; the present project focuses on visualization.

1.4 Process Flow of the System

COVID-19 Raw Data	→	**Data Cleaning**	→	**Data Transformation**
↓		↓		↓
Data Analysis	→	**KPI Calculation**	→	**Visualization Layer**
		↓		↓
Interactive Dashboard	→	**Filters & Slicers**	→	**Trends & Insights**

2. Dataset

The dataset is a structured tabular collection of COVID-19 observations. The exact number of records and final column list depend on the dataset selected for implementation and must be taken directly from that dataset.

Dataset Detail	Description
Source of data	COVID-19 public-health dataset selected for the final project implementation. The exact source should be stated in the final report.
Number of records / samples	Dataset-dependent. The exact count should be taken from the final CSV/Excel file.
Features / attributes	Typical fields include Date, Country/Region or State, Confirmed Cases, New Cases, Deaths, New Deaths, Recoveries, Active Cases, Testing and Vaccination fields when available.
Training and testing split	Not applicable to the present descriptive visualization project because a predictive ML model is not required. A split can be introduced if a future prediction module is added.
Data format / type	Structured tabular data, commonly CSV or Excel. Numerical fields contain counts or measurements; date and location fields provide time and geographical dimensions.

2.1 Features / Attributes (Columns)

Attribute	Purpose
Date	Time-based trend analysis and filtering.
Country / Region / State	Geographical comparison and filtering.
Confirmed Cases	Measures reported COVID-19 cases.
Deaths	Supports mortality and outcome analysis.
Recoveries	Supports recovery analysis where available.
Active Cases	Represents ongoing reported cases where available or derivable.
Testing / Vaccination	Supports additional public-health analysis when these fields exist.

3. Tools and Technologies Used

- **Python & Pandas:** Data loading, cleaning, transformation and exploratory analysis.
- **Excel / CSV:** Dataset source or intermediate validation format.
- **Power BI:** Interactive dashboard, KPI cards, charts, filters and slicers.
- **DAX:** Calculated measures where Power BI is used.
- **Power Query:** Importing and shaping data inside the BI platform.

4. Pseudo Code

```
BEGIN COVID19_DataPipeline
raw_data ← LOAD(covid19_dataset.csv)
CHECK data types, ranges and missing values
REMOVE_DUPLICATES(raw_data)
HANDLE_MISSING_VALUES(raw_data)
STANDARDIZE_LOCATION_NAMES(raw_data)
clean_data ← TRANSFORM(raw_data)
EXTRACT day, month, quarter, year
CALCULATE supported COVID-19 KPIs
CREATE trend and comparison visualizations
BUILD interactive dashboard
CROSS_CHECK dashboard values with clean_data
END COVID19_DataPipeline
```

5. Expected Output

An interactive COVID-19 dashboard that presents key indicators, time-based trends, geographical comparisons and other available analytical views. Users can apply filters and quickly interpret patterns from the cleaned dataset. The project demonstrates the application of data preparation, descriptive analytics and Business Intelligence visualization in the Computer Science and Engineering (Artificial Intelligence and Machine Learning) domain.